B25-L6 lab: Build a small transformer from scratch and prove each part works

Platform: Laptop

Work in `labs/transformer/` in this repository. You build attention, multi-head attention, sinusoidal positions and a pre-norm block in PyTorch, then train a small classifier on a synthetic task: does a sequence start and end with the same token? The task needs positions, and the tests check your attention against PyTorch's own, and that padding and the future cannot leak. You need PyTorch 2.15 on the laptop. When it trains, push the lab to your own GitHub repository with a written explanation: it is the roadmap's evidence for month 9, a transformer built from scratch.

1. Run `cd labs/transformer`, create a virtual environment, then `pip install -e .` and `pip install pytest`. Run `pytest`: most tests fail with NotImplementedError. Read `README.md` and every file in `src/tiny_transformer/`.
2. In `src/tiny_transformer/layers.py`, write `attention` (lessons B25-L3 and B25-L4). Run `pytest tests/test_layers.py -k attention`: it compares your output with `F.scaled_dot_product_attention`.
3. Write `sinusoidal_positions` and `MultiHeadSelfAttention` (lesson B25-L5), then `Block` (lesson B25-L6). Run `pytest tests/test_layers.py`.
4. Write `Classifier` in `src/tiny_transformer/model.py`: token embeddings plus positions, blocks, the mean over real tokens, and a linear head. Run `pytest`: every test passes.
5. Run `python -m tiny_transformer train`, then `python -m tiny_transformer train --no-positions`. Compare the accuracy and explain the gap.
6. Write your README: each part, the test that proves it, the two runs, and a diagram of one block. Push the folder to a new GitHub repository.

Notes